

# DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning

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## Abstract

General reasoning represents a long-standing and formidable challenge in artificial intelligence. Recent breakthroughs, exemplified by large language models (LLMs) (Brown et al., 2020; OpenAI, 2023) and chain-of-thought prompting (Wei et al., 2022b), have achieved considerable success on foundational reasoning tasks. However, this success is heavily contingent upon extensive human-annotated demonstrations, and models’ capabilities are still insufficient for more complex problems. Here we show that the reasoning abilities of LLMs can be incentivized through pure reinforcement learning (RL), obviating the need for human-labeled reasoning trajectories. The proposed RL framework facilitates the emergent development of advanced reasoning patterns, such as self-reflection, verification, and dynamic strategy adaptation. Consequently, the trained model achieves superior performance on verifiable tasks such as mathematics, coding competitions, and STEM fields, surpassing its counterparts trained via conventional supervised learning on human demonstrations. Moreover, the emergent reasoning patterns exhibited by these large-scale models can be systematically harnessed to guide and enhance the reasoning capabilities of smaller models.

## 1. Introduction

Reasoning capability, the cornerstone of human intelligence, enables complex cognitive tasks ranging from mathematical problem-solving to logical deduction and programming. Recent advances in artificial intelligence have demonstrated that large language models (LLMs) can exhibit emergent behaviors, including reasoning abilities, when scaled to a sufficient size (Kaplan et al., 2020; Wei et al., 2022a). However, achieving such capabilities in pre-training typically demands substantial computational resources. In parallel, a complementary line of research has demonstrated that large language models can be effectively augmented through chain-of-thought (CoT) prompting. This technique, which involves either providing carefully designed few-shot examples or using minimalistic prompts such as “Let’s think step by step” (Kojima et al., 2022; Wei et al., 2022b), enables models to produce intermediate reasoning steps, thereby substantially enhancing their performance on complex tasks. Similarly, further performance gains have been observed when models learn high-quality, multi-step reasoning trajectories during the post-training phase (Chung et al., 2024; OpenAI, 2023). Despite their effectiveness, these approaches exhibit notable limitations. Their dependence on human-annotated reasoning traces hinders scalability and introduces cognitive biases. Furthermore, by constraining models to replicate human thought processes, their performance is inherently capped by the human-

provided exemplars, which prevents the exploration of superior, non-human-like reasoning pathways.

To tackle these issues, we aim to explore the potential of LLMs for developing reasoning abilities through self-evolution in an RL framework, with minimal reliance on human labeling efforts. Specifically, we build upon DeepSeek-V3-Base (DeepSeek-AI, 2024b) and employ Group Relative Policy Optimization (GRPO) (Shao et al., 2024) as our RL framework. The reward signal is solely based on the correctness of final predictions against ground-truth answers, without imposing constraints on the reasoning process itself. Notably, we bypass the conventional supervised fine-tuning (SFT) phase before RL training. This design choice stems from our hypothesis that human-defined reasoning patterns may limit model exploration, whereas unrestricted RL training can better incentivize the emergence of novel reasoning capabilities in LLMs. Through this process, detailed in Section 2, our model (referred to as DeepSeek-R1-Zero) naturally developed diverse and sophisticated reasoning behaviors. In solving reasoning problems, the model exhibits a tendency to generate longer responses, incorporating verification, reflection, and the exploration of alternative approaches within each response. Although we do not explicitly teach the model how to reason, it successfully learns improved reasoning strategies through reinforcement learning.

Although DeepSeek-R1-Zero demonstrates excellent reasoning capabilities, it faces challenges such as poor readability and language mixing, occasionally combining English and Chinese within a single chain-of-thought response. Furthermore, the rule-based RL training stage of DeepSeek-R1-Zero is narrowly focused on reasoning tasks, resulting in limited performance in broader areas such as writing and open-domain question answering. To address these challenges, we introduce DeepSeek-R1, a model trained through a multi-stage learning framework that integrates rejection sampling, reinforcement learning, and supervised fine-tuning, detailed in Section 3. This training pipeline enables DeepSeek-R1 to inherit the reasoning capabilities of its predecessor, DeepSeek-R1-Zero, while aligning model behavior with human preferences through additional non-reasoning data.

To enable broader access to powerful AI at a lower energy cost, we have distilled several smaller models and made them publicly available. These distilled models exhibit strong reasoning capabilities, surpassing the performance of their original instruction-tuned counterparts. We believe that these instruction-tuned versions will also significantly contribute to the research community by providing a valuable resource for understanding the mechanisms underlying long chain-of-thought (CoT) reasoning models and for fostering the development of more powerful reasoning models. We release DeepSeek-R1 series models to the public at <https://huggingface.co/deepseek-ai>.

## **2. DeepSeek-R1-Zero**

We begin by elaborating on the training of DeepSeek-R1-Zero, which relies exclusively on reinforcement learning without supervised fine-tuning. To facilitate large-scale RL efficiency, we adopt Group Relative Policy Optimization (GRPO) (Shao et al., 2024).

### **2.1. Group Relative Policy Optimization**

GRPO (Shao et al., 2024) is the reinforcement learning algorithm that we adopt to train DeepSeek-R1-Zero and DeepSeek-R1. It was originally proposed to simplify the training process and reduce the resource consumption of Proximal Policy Optimization (PPO) (Schulman et al., 2017), which is widely used in the RL stage of LLMs (Ouyang et al., 2022).

For each question  $q$ , GRPO samples a group of outputs  $\{o_1, o_2, \dots, o_G\}$  from the old policy  $\pi_{\theta_{old}}$  and then optimizes the policy model  $\pi_{\theta}$  by maximizing the following objective:

$$\mathcal{J}_{GRPO}(\theta) = \mathbb{E}[q \sim P(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{old}}(O|q)] \frac{1}{G} \sum_{i=1}^G \left( \min \left( \frac{\pi_{\theta}(o_i|q)}{\pi_{\theta_{old}}(o_i|q)} A_i, \text{clip} \left( \frac{\pi_{\theta}(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}, 1 - \varepsilon, 1 + \varepsilon \right) A_i \right) - \beta \mathbb{D}_{KL}(\pi_{\theta} || \pi_{ref}) \right), \quad (1)$$

$$\mathbb{D}_{KL}(\pi_{\theta} || \pi_{ref}) = \frac{\pi_{ref}(o_i|q)}{\pi_{\theta}(o_i|q)} - \log \frac{\pi_{ref}(o_i|q)}{\pi_{\theta}(o_i|q)} - 1, \quad (2)$$

where  $\pi_{ref}$  is a reference policy,  $\varepsilon$  and  $\beta$  are hyper-parameters, and  $A_i$  is the advantage, computed using a group of rewards  $\{r_1, r_2, \dots, r_G\}$  corresponding to the outputs within each group:

$$A_i = \frac{r_i - \text{mean}(\{r_1, r_2, \dots, r_G\})}{\text{std}(\{r_1, r_2, \dots, r_G\})}. \quad (3)$$

We give a comparison of GRPO and PPO in Supplementary A.3. To train DeepSeek-R1-Zero, we set the learning rate to 3e-6, the KL coefficient to 0.001, and the sampling temperature to 1 for rollout. For each question, we sample 16 outputs with a maximum length of 32,768 tokens before the 8.2k step and 65,536 tokens afterward. As a result, both the performance and response length of DeepSeek-R1-Zero exhibit a significant jump at the 8.2k step, with training continuing for a total of 10,400 steps, corresponding to 1.6 training epochs. Each training step consists of 32 unique questions, resulting in a training batch size of 512. Every 400 steps, we replace the reference model with the latest policy model. To accelerate training, each rollout generates 8,192 outputs, which are randomly split into 16 mini-batches and trained for only a single inner epoch.

Table 1 | Template for DeepSeek-R1-Zero. **prompt** will be replaced with the specific reasoning question during training.

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A conversation between User and Assistant. The user asks a question, and the Assistant solves it. The assistant first thinks about the reasoning process in the mind and then provides the user with the answer. The reasoning process and answer are enclosed within `<think>...</think>` and `<answer>...</answer>` tags, respectively, i.e., `<think>` reasoning process here `</think>` `<answer>` answer here `</answer>`. User: **prompt**. Assistant:

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Our high-performance RL infrastructure is described in Supplementary B.1, ensuring scalable and efficient training.

## 2.2. Reward Design

The reward is the source of the training signal, which decides the direction of RL optimization. For DeepSeek-R1-Zero, we employ rule-based rewards to deliver precise feedback for data in mathematical, coding, and logical reasoning domains. Our rule-based reward system mainly consists of two types of rewards: accuracy rewards and format rewards.

**Accuracy rewards** evaluate whether the response is correct. For example, in the case of math problems with deterministic results, the model is required to provide the final answer in a specified format (e.g., within a box), enabling reliable rule-based verification of correctness. Similarly, for code competition prompts, a compiler can be utilized to evaluate the model's